ICDAR 2026

Patram-Bench

A Multi-task, Multi-domain & Multi-lingual Benchmark for Indian Document Image Understanding

Anirudh Srinivasan[1,*] **Pratyush Jena[1,2,*]** Arya Topale[2,†] Venkat Venna[1,†] Ravi Kiran Sarvadevabhatla[1,2]

[1] BharatGen [2] IIIT Hyderabad

The world's largest Indic multilingual document understanding benchmark

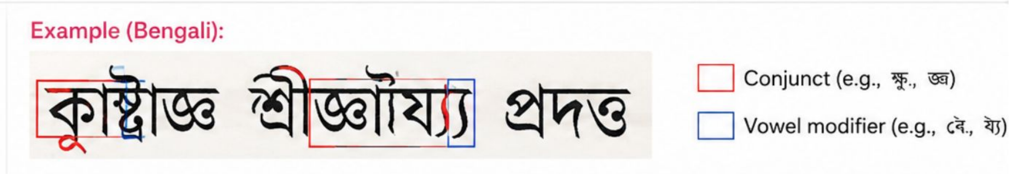
52K	2.4M	23	52	13	71	29
DOCUMENTS	QA ANNOTATIONS	LANGUAGES	TASKS	DOMAINS	DOC CLASSES	VLMs EVALUATED

THE CHALLENGE

India has 1.4 billion people, 22 official languages and dozens of scripts — yet document understanding benchmarks remain Western- and English-centric.

- Connected headlines — Shirorekha joins characters across a whole word.

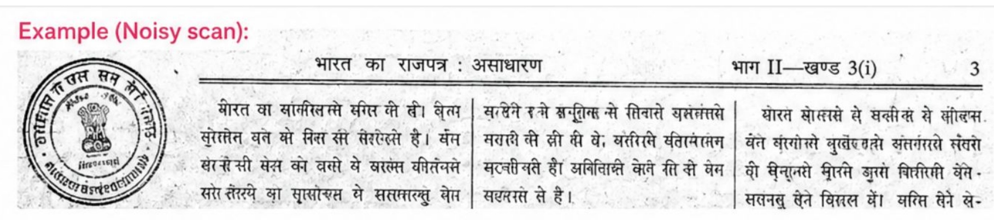
- Complex ligatures — Conjuncts and spatially disjoint vowel modifiers.



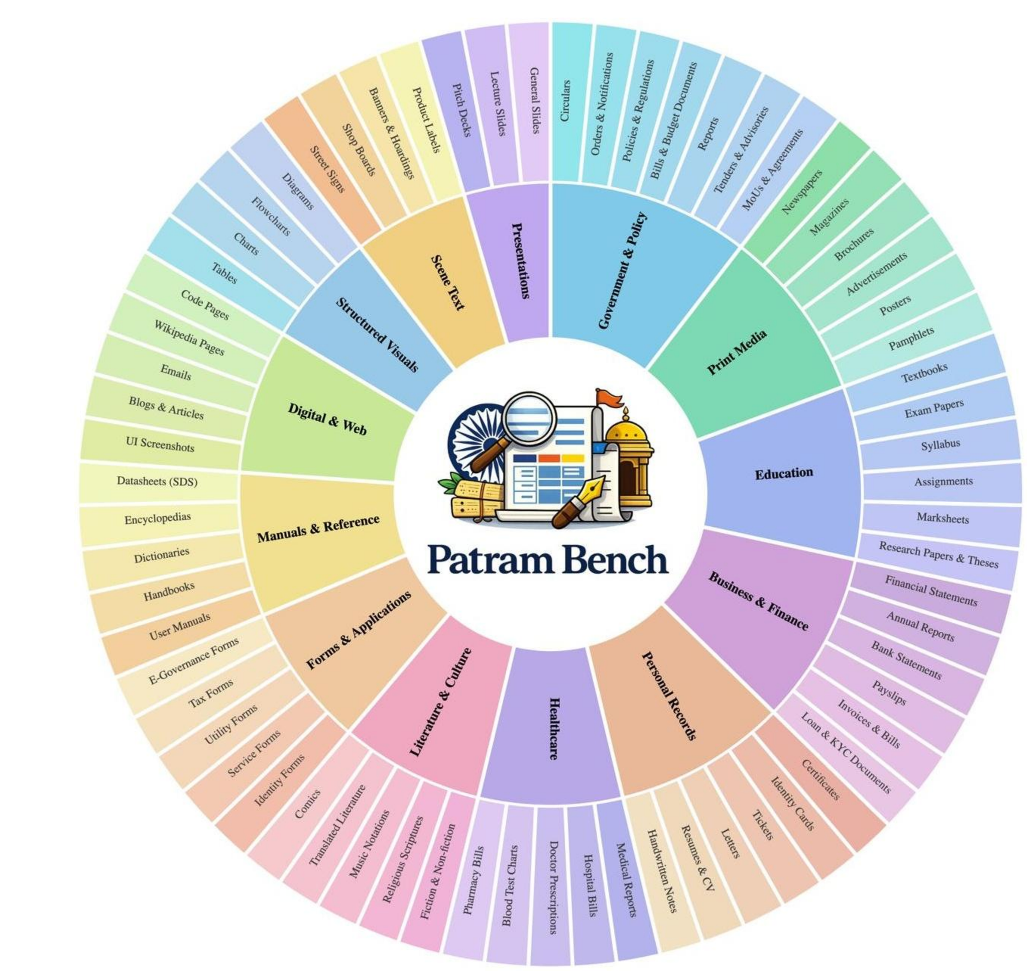
- Order mismatch — Rendering order diverges from Unicode order.

- Script mixing — Multiple scripts inside a single page.

- Real-world noise — Degraded scans, seals, dense multi-column layouts.

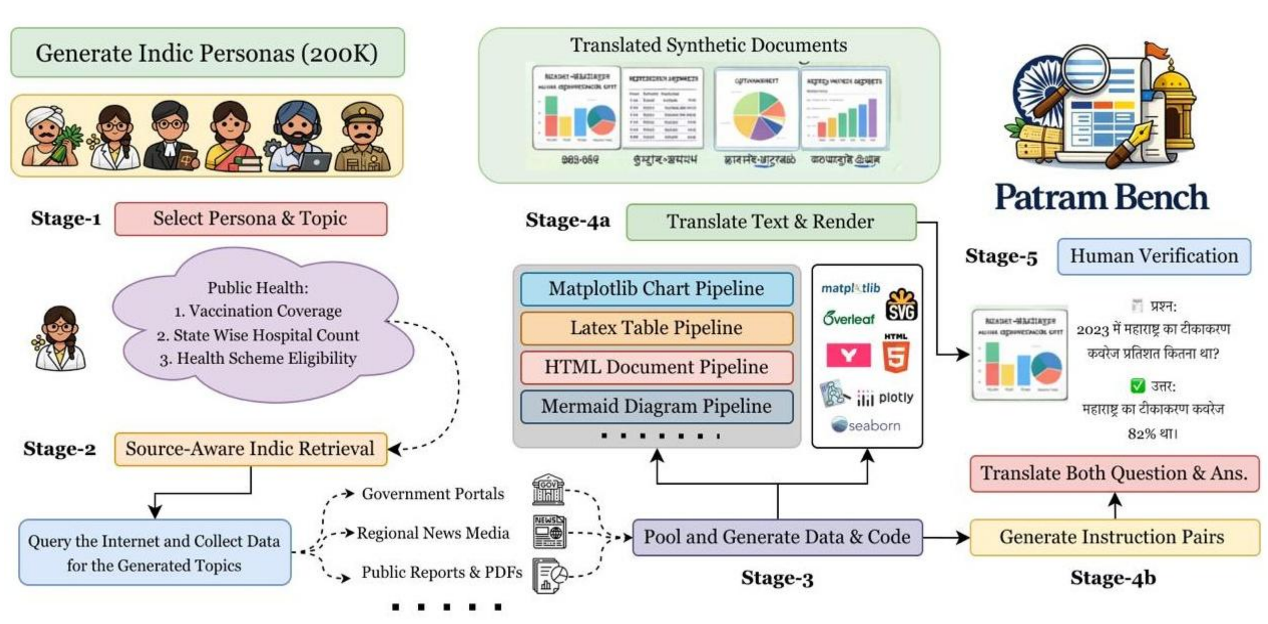


13 DOMAINS · 71 CLASSES



Inner ring: 13 high-level domains. Outer ring: 71 fine-grained document classes.

PATRAM-SYN SYNTHESIS



200k+ Indic personas → source-aware retrieval → Matplotlib / LaTeX / HTML / Mermaid rendering → 22 languages → expert human verification.

52 TASKS · 8 FAMILIES

PERCEPTION-BASED TASKS

Text Recognition
Scene text · Handwritten · Full-page · Line · Word OCR

Document Segmentation
Word · Line · Paragraph · Layout detection

Document Parsing
Document → Markdown · HTML · LaTeX

Structured Extraction
Chart → JSON / DataFrame · Table → HTML / LaTeX · Table-cell localisation · Diagram → JSON · Form parsing

REASONING-BASED TASKS

Document VQA
Extractive · Abstractive · Summarisation · Key information extraction · Layout-based · Language / script ID · Quality assessment

Knowledge & Reasoning
Multi-hop · Arithmetic · Logical · Ambiguous · Adversarial · Analogy · Counting · Compound · Science · Comparison

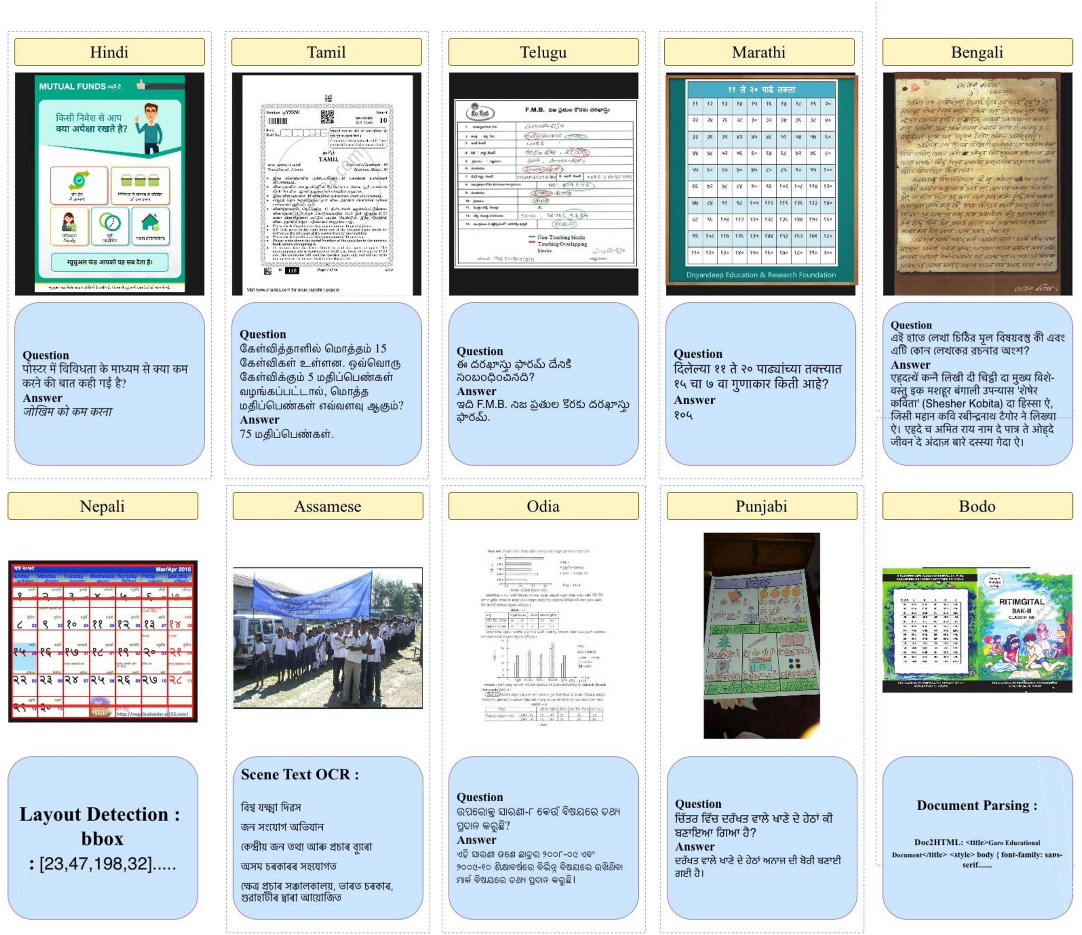
Structured Reasoning
ChartVQA · TableVQA · Table recognition · DiagramVQA · FormVQA

Linguistic Robustness
Indic VQA · Cross-lingual · Transliterated · Code-mixed

LARGEST OF ITS KIND

Benchmark	Indic	#Lang	#Tasks	#Docs	#Dom	#Cls	#Ann
MT-VQA	No	9	1	8.8k	–	–	21.8k
XFUND-test	No	7	1	455	1	–	97k+
OmniDocBench	No	3	5	981	9	19	100k+
OCRBench v2	No	2	23	9.5k	–	–	10k
CC-OCR	No	10	4	7.1k	7	18	7.1k
KITAB-Bench	No	1	8	8.8k	9	36	8.8k
IndicDLP-test	Yes	12	1	15k+	12	42	185.6k
IndicVisionBench	Yes	10	3	5k	–	–	37k+
Patram-Bench (Ours)	**Yes**	**23**	**52**	**53k**	**13**	**71**	**2.4M**

MULTILINGUAL BY DESIGN



Perception and reasoning tasks across high- and low-resource languages — Hindi, Tamil, Telugu, Marathi, Bengali, Nepali, Assamese, Odia, Punjabi, Bodo.

ANNOTATION & GOLD SET

Gemini 3 Pro — seeds question–answer pairs › **LLM QA/QC** — flags hallucination & grounding errors › **14 annotators** — review a stratified subset › **51.3k gold QA** — human-validated benchmark set

29 SYSTEMS EVALUATED

Closed-source LLMs	Gemini 3 Pro · GPT-5 · Claude 4 Sonnet
OCR APIs	Sarvam Vision · Chitrapathak
Open large	Qwen3-VL 235B · InternVL3.5 241B · GLM-4.5V · LLaMA 4 Scout
Open medium	Qwen3-VL 30B / 32B · InternVL3.5 30B / 38B · Gemma 3 27B · DeepSeek-VL2 · LLaVA 1.6
Open small	Qwen3-VL 8B · Qwen2.5 7B · InternVL 8B · Molmo2 8B · Gemma 3 12B · Pixtral 12B · MiniCPM-V 4.5
Specialist document models	Surya · Marker · Chandra · DeepSeek-OCR · dots.ocr · DocLayout-YOLO · IndicDLP · Hi-SAM · DocTr

TWO NEW METRICS

DUCLI — Document Understanding Cross-Lingual Index

$$\mathrm{DUCLI} = \frac{1}{4} \cdot \frac{S(I,I) + S(E,I) + S(I,E) + S(I_H,I_L)}{S(E,E)}$$

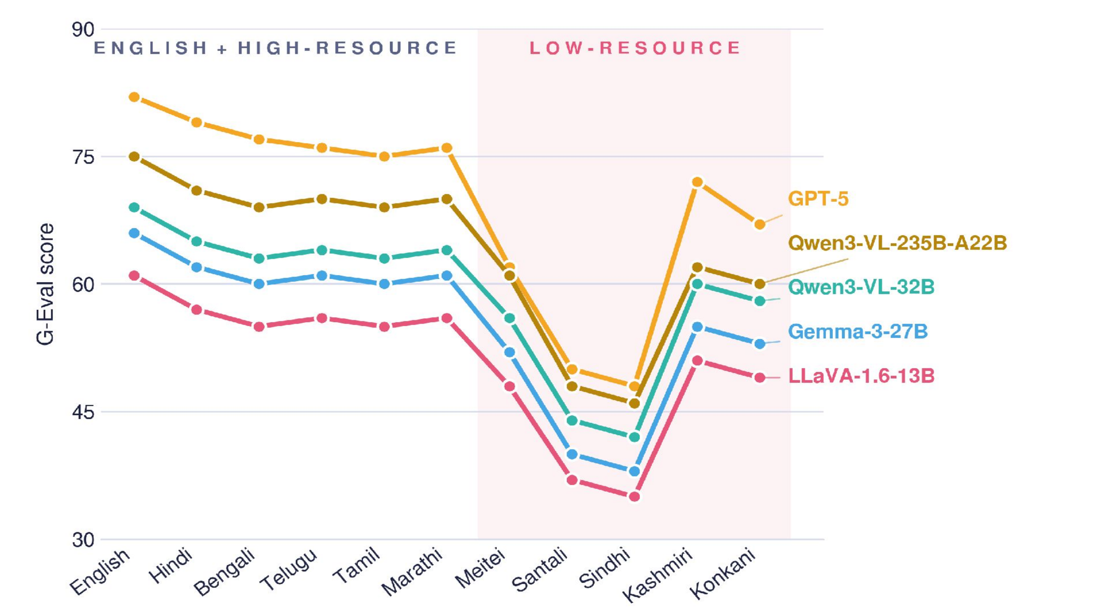
Performance retained across four transfer settings, relative to an English–English baseline.

SRS — Script Robustness Score

$$\mathrm{SRS} = \frac{1}{2}\left(\frac{S_{cm}}{S_{pure}} + \frac{S_{tl}}{S_{pure}}\right)$$

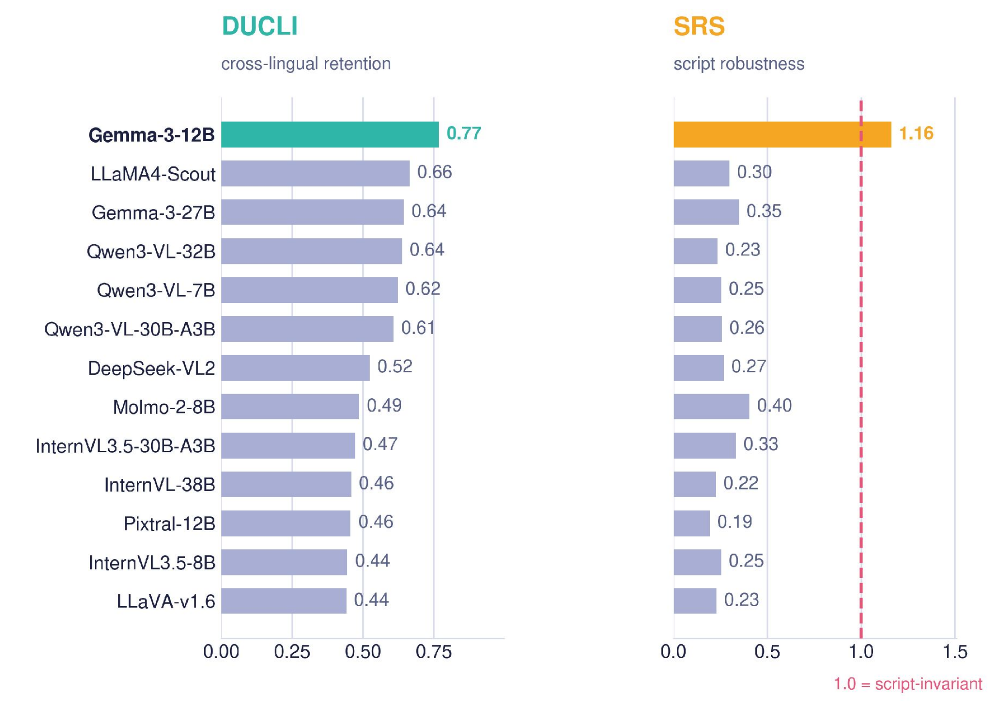
Resilience to intra-document script perturbation — code-mixed and transliterated input. 1.0 = script-invariant.

LANGUAGE ROBUSTNESS



G-Eval across English, five high-resource and five low-resource Indic languages. The slope from left to right is the robustness story.

DUCLI & SRS



Open models ordered by DUCLI. Gemma-3-12B leads both metrics despite its modest scale; its SRS above 1.0 means perturbed input scores no worse than native.

KEY FINDINGS

1. **Task sensitivity**
Gemini 3 Pro leads overall; Claude 4 Sonnet wins text-heavy parsing. For layout, specialists (IndicDLP, Surya, Hi-SAM) still beat general VLMs outright.

2. **The low-resource cliff**
The proprietary–open gap widens as resources decline. Qwen3-VL leads on high-resource languages but falls away on low-resource ones, where Gemma-3-27B takes over.

3. **Script diversity, not scale**
Gemma-3-12B tops the open set on both DUCLI (0.767) and SRS (1.16) despite its modest size. Scores decline from English to cross-lingual to cross-Indic transfer.

4. **Bigger models read the page less**
Qwen3-VL-235B and InternVL3.5-241B lean hardest on language priors — answering plausibly with the image removed. The most visually grounded open models are mid-sized.

5. **Scale is not the answer**
Cross-script robustness tracks script diversity and cross-lingual supervision not parameter count. Multilingual document intelligence needs structurally grounded, visually faithful models.

FIND EVERYTHING ONLINE

Project page patrambench.github.io

Project page